LBW BALL-TRACKING REPORT

Shamim_LBW.mp4 · left-handed batter · on-field OUT · shot offered

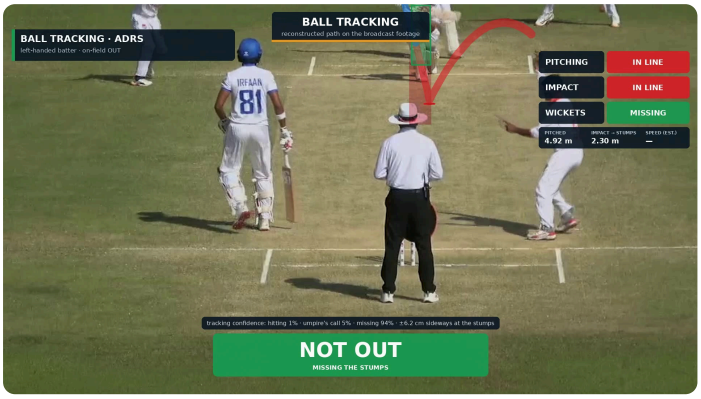
Ball tracking · ADRS
21 Oct 2026, 17:02

NOT OUT

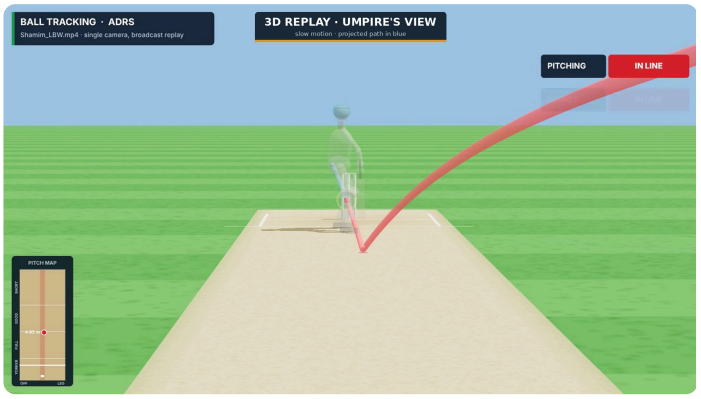
missing the stumps

LBW protocol

1	**Pitching**	IN LINE	Bounced 4.92 m from the stumps, -11 cm across Outside leg only if more than half the ball pitched outside the line of leg stump.
2	**Impact**	IN LINE	Hit the pad 2.30 m from the stumps, +6 cm across, 46 cm high In line if at least half the ball is in line; umpire's call if 1–50%; outside if none.
3	**Wickets**	MISSING	Projected to cross the stumps +19.6 cm across, 73 cm high Zone = outer edges of the stumps up to the top of the bails. Hitting if the ball's centre is inside; umpire's call if only part of the ball is; missing if none of it is.
4	**Distance**	NO RULE	Impact→stumps 2.30 m, pitch→impact 2.62 m Impact 3 m or more from the stumps, or 2.5–3 m with pitch→impact under 40 cm → umpire's call.



Ball tracking drawn on the broadcast footage: path (red), bounce, projection (blue), stumps zone

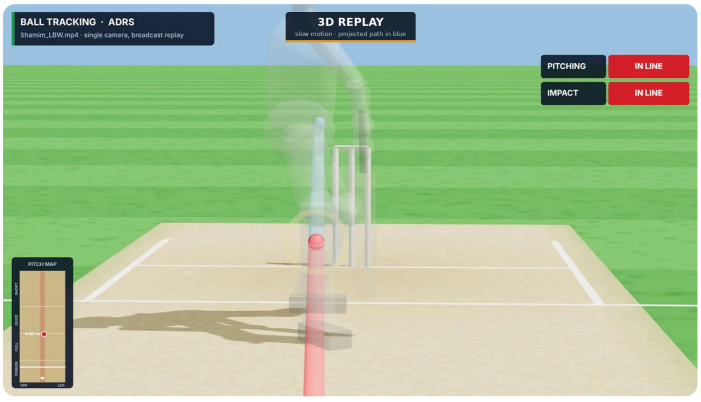


3D replay · umpire's view

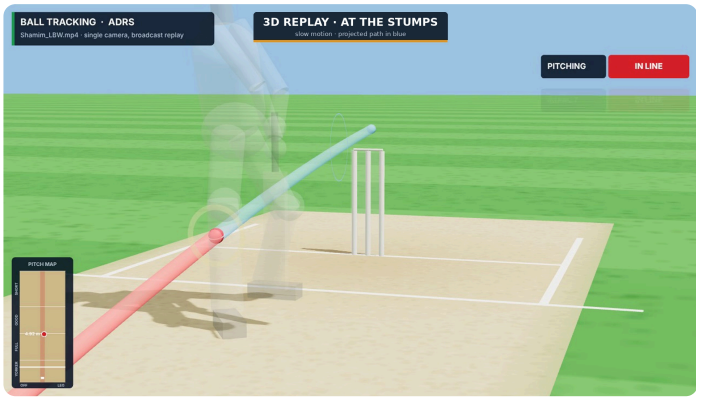
Measurements

Bounce point (across, from stumps)	**-0.111 m, 4.92 m**	±1.0 cm, ±28 cm
Impact (across, from stumps, height)	**+0.055 m, 2.30 m, 0.46 m**	
At the stumps (across, height)	**+0.196 m, 0.733 m**	±6.2 cm, ±23.0 cm
Plausible paths	**hitting 1% · umpire's call 5% · missing 94%**	from the measurement uncertainty
Speed	**not determined**	
Ball positions used	**21 of 21 (4 after the bounce)**	fit 0.67 px rms
Camera calibration	**crease corners found automatically**	1.75 px rms · camera [-0.32, 77.69, 6.98] m
Front foot at impact	**found automatically**	pixel [1150, 216]

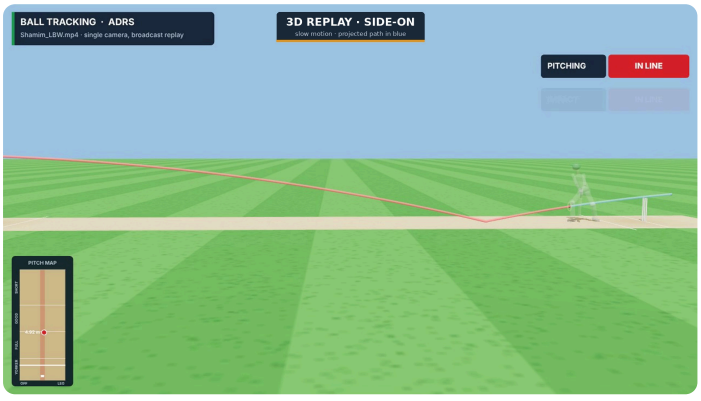
3D replay



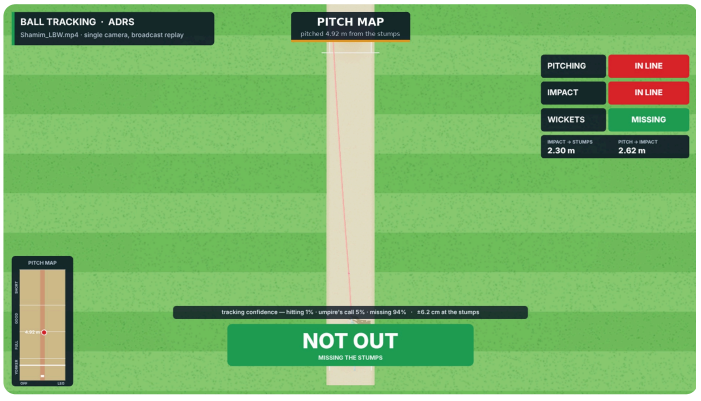
Main view at the decision



At the stumps: the projected path against the wicket zone



Side-on: height of the ball through bounce, impact and the stumps



Pitch map: where the ball bounced

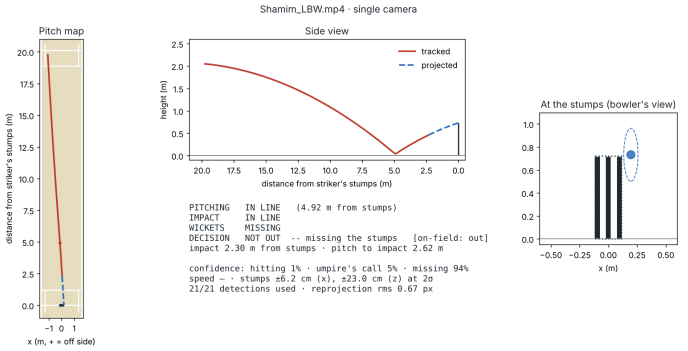
How it was measured



What the system found by itself: the crease corners used to calibrate the camera (left), and the ball and front foot at impact (right)



Calibration check: the pitch model (magenta) and stumps (red) drawn over the footage



Technical view: path from above, from the side and end-on, with the projection and its uncertainty

- **Camera.** The broadcast camera's position, direction and zoom are solved from the painted creases at both ends.
- **Ball.** The ball is found in every frame of the replay as a small dark, red-brown blob that moves along a smooth path; frames repeated by the slow motion are merged.
- **Path.** A physical path is fitted: two arcs under gravity joined by a bounce on the pitch. Broadcast slow motion has no usable timing, so only the order of the ball positions is used, and speed comes out as an estimate.
- **Impact.** The front foot's position at impact fixes how far down the pitch the ball hit the pad.
- **Decision.** The LBW protocol is applied to the best-estimate path. The percentages show how the call could change within the measurement uncertainty.

Limits

- One camera behind the bowler measures sideways movement well but distance down the pitch less well, so its projection is wider than a multi-camera system's.
- The footage must be from behind the bowler with no zoom or pan during the delivery, and a red ball.
- Check the playing conditions (distance rules, wicket zone) against the competition's own before match use.